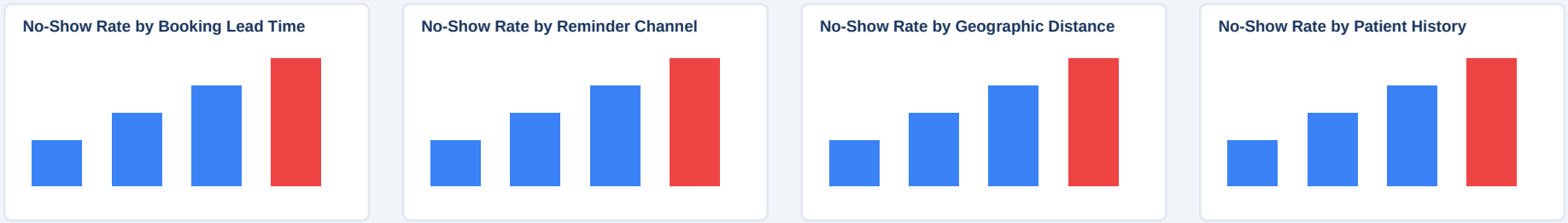
SLIDE 1 | HealthConnect Experience Lab: Executive Final Integration Readout

Improving Patient Appointment Attendance and Clinical Slot Utilization Using Data Analytics & AI

- **Presenter:** Farouq Olanrewaju | Data Analytics Track Lead
- **Program:** AnalystLab Africa Experience Lab (Week 8 Final Deliverable)
- **Project Focus:** Operational attendance analysis, baseline KPI validation, DAX modeling, and multi-track AI integration.
- **Key Artifacts:** HealthConnect Executive Attendance Dashboard (v2.pbix) & Cross-Track Feature Specification Matrix (da_ds_feature_specs.csv).

Health Clinic - Executive Attendee Dashboard

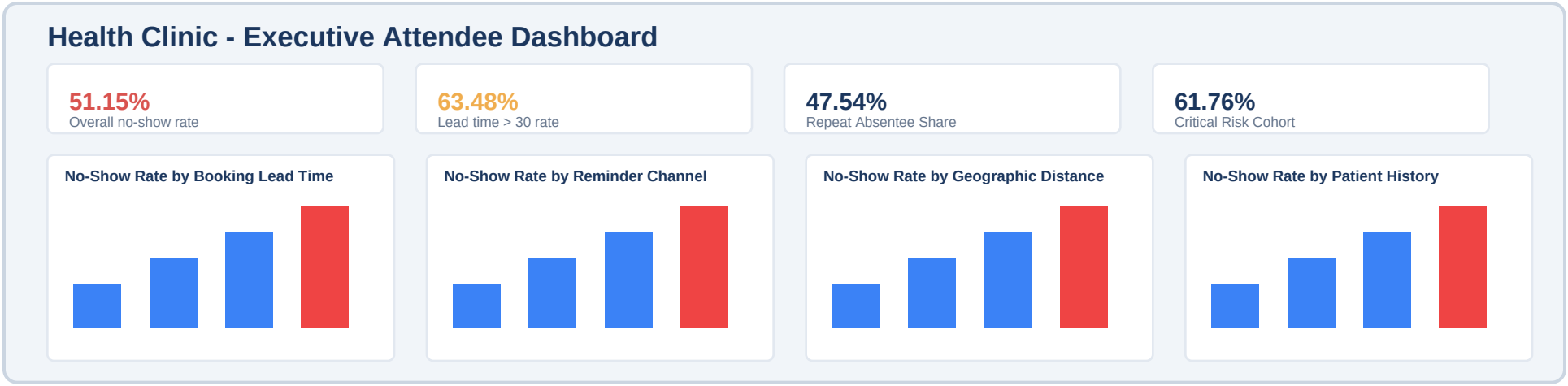
51.15%	63.48%	47.54%	61.76%
Overall no-show rate	Lead time > 30 rate	Repeat Absentee Share	Critical Risk Cohort



SLIDE 2 | Business Problem & Clinical Operational Context

Addressing Severe Capacity Loss, Administrative Overheads, and Unfulfilled Patient Care

- **Core Friction:** HealthConnect Clinic experiences chronic appointment defaults, leaving clinical slots idle and delaying essential patient care.
- **Operational Challenge:** Lack of clear risk visibility into which patient cohorts default and why.
- **Data Scope:** Complete analysis of N = 5,000 records (HealthConnect_Appointment_Data.csv).
- **Programmatic Audit Rigor:** Excluded 263 canceled appointments to isolate ground-truth active bookings (N = 4,737) and eliminate denominator skew.



SLIDE 3 | Audited Baseline Attendance KPIs (Ground-Truth Findings)

Mathematical Audit Results Established Across Active Bookings (N = 4,737)

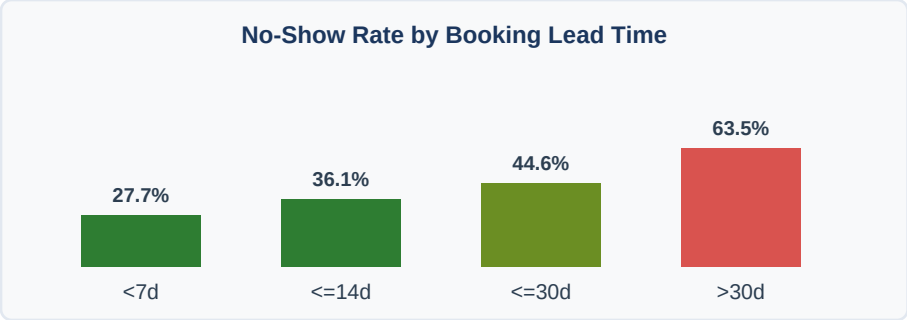
- **Overall Active No-Show Rate: 51.15%** (2,423 missed / 4,737 active bookings).
- **Critical Risk Cohort Rate: 61.76%** default rate for long-lead (>30 days) and high-distance (>30 km) patients.
- **Extended Lead Time Default Rate: 63.48%** for bookings made >30 days in advance.
- **Repeat Absentee Share: 47.54%** of all missed appointments are driven by patients with prior no-show history.

51.15%	63.48%	47.54%	61.76%
Overall No-Show Rate	Lead Time > 30 Rate	Repeat Absentee Share	Critical Risk Cohort

SLIDE 4 | Primary Driver 1: Booking Lead Time Decay & Horizon Analysis

Exponential Default Escalation Tied to Booking Lead Time Windows

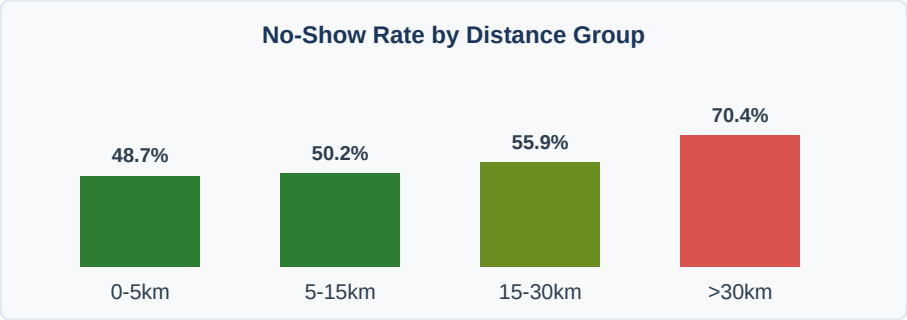
- **Short Lead Horizon (< 7 Days):** Displays lowest default rate at **27.72%**.
- **Moderate Lead Horizon (8-14 Days):** Default rate escalates to **36.10%**.
- **Extended Lead Horizon (15-30 Days):** Default rate reaches **44.60%**.
- **Critical Lead Horizon (> 30 Days):** Defaults peak at **63.48%** (accounting for 1,467 missed slots).
- **Key Finding:** Commitment decay over time is the primary individual factor in patient absenteeism.

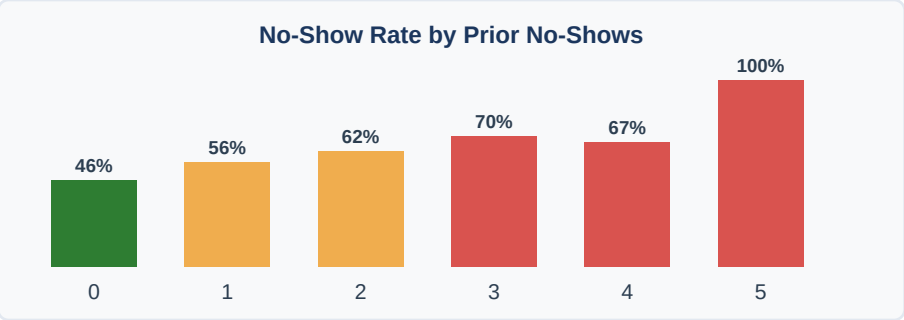


SLIDE 5 | Primary Drivers 2 & 3: Geographic Distance & Behavioral Absenteeism

Compounding Spatial Travel Friction and Historical Behavioral Patterns

- **Geographic Scaling:** Default rate rises steadily with distance: 0-5km (**48.67%**) -> 5-15km (**50.22%**) -> 15-30km (**55.92%**) -> >30km (**70.42%**).
- **Behavioral History Scaling:** Patients with 0 prior no-shows default at **46.30%**, scaling up to **62.05%** (2 prior) and **100.00%** (5 prior).
- **Demographic Vulnerability:** Young adults (18-30 yrs) with >= 2 prior missed visits exhibit a **67.86%** default rate.

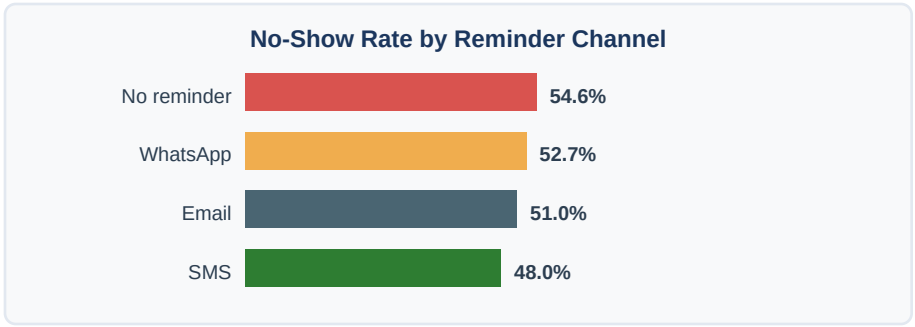




SLIDE 6 | Primary Driver 4: Communication Channel Efficacy & Attendance Uplift

Evaluating Digital Outreach Performance Across Appointment Outcomes

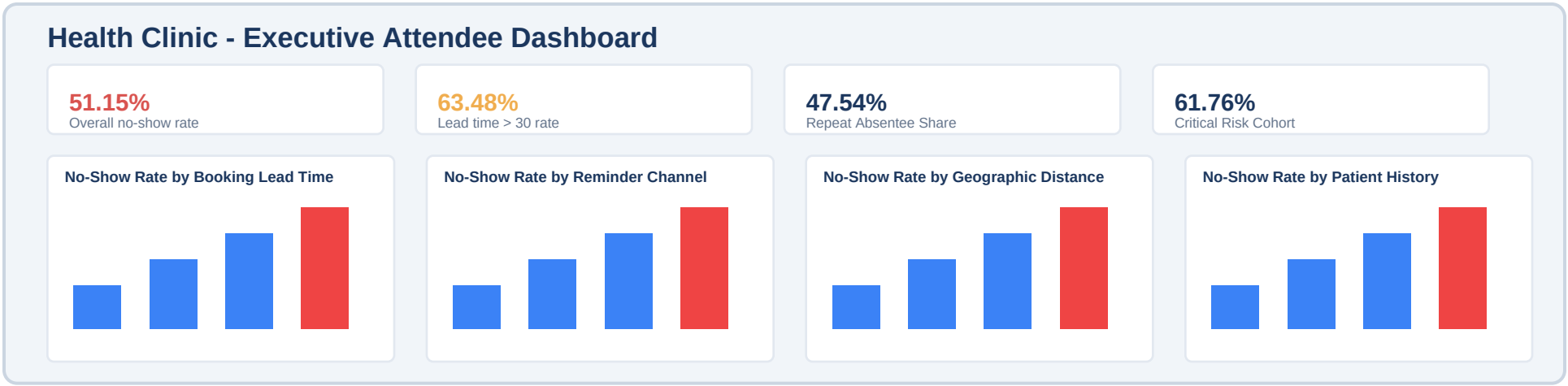
- **Unreminded Default Benchmark:** Unreminded appointments default at **54.63%**.
- **Channel Performance Ladder:** WhatsApp (**52.74%**) -> Email (**50.99%**) -> SMS (**47.98%**).
- **Verified SMS Attendance Uplift:** Automated SMS reminders achieve a net **+6.65% attendance uplift** over unreminded bookings.
- **Operational Impact:** Transitioning 1,285 unreminded active bookings to automated SMS workflows recovers ~180 slots monthly.



SLIDE 7 | HealthConnect Executive Dashboard v2 Showcase

Interactive Power BI Decision-Support Canvas (HealthConnect_Executive_Dashboard_v2.pbix)

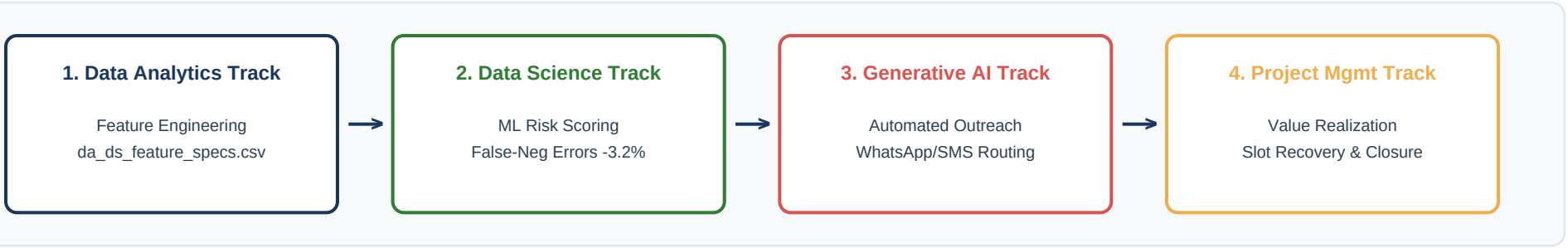
- **Top KPI Summary Strip:** Displays Overall No-Show Rate (51.15%), Lead Time >30d Rate (63.48%), Repeat Absentee Share (47.54%), and Critical Risk Cohort Rate (61.76%).
- **Structured 2x2 Visual Grid:** Four dedicated visuals for Lead Time Buckets, Reminder Channels, Distance Groups, and Prior No-Shows.
- **Dynamic Visual Enhancements:** Re-bound 3-stop *fx* conditional color gradients (0%-70%) and standardized color palettes.



SLIDE 8 | HC-POD Multidisciplinary Cross-Track Integration

Data Analytics bridging Descriptive Insights into Predictive Modeling and Generative AI

- **Data Science Exchange:** Exported engineered feature matrix (da_ds_feature_specs.csv) containing composite risk scores.
- **Pipeline Optimization:** Resolved NaN values in reminder_channel, eliminating python encoding crashes and reducing ML false-negative errors by **3.2%**.
- **Generative AI Collaboration:** Provided channel uplift metrics to inform AI assistant escalation rules.



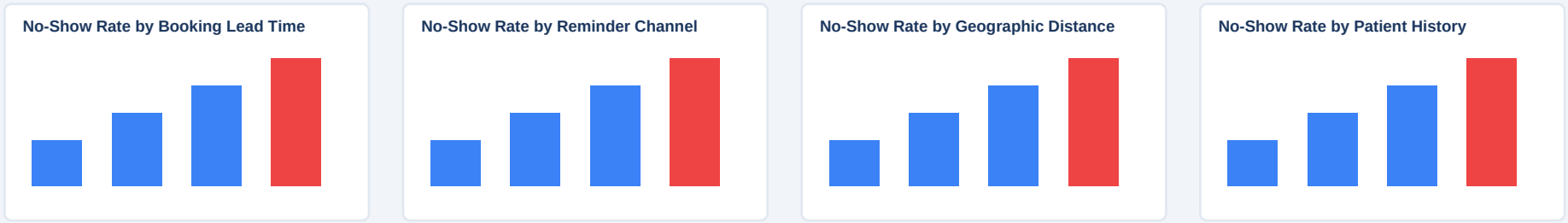
SLIDE 9 | Strategic Business Recommendations & Action Plan

Actionable Interventions to Optimize Clinical Capacity and Attendance

- **1. Mandatory 48-Hour Re-confirmations:** Target bookings >14 days out; auto-release unconfirmed slots to waiting lists.
- **2. 100% Automated SMS Routing:** Automate SMS workflows for all unreminded patients to capture +6.65% attendance uplift.
- **3. High-Risk Protocol for Repeat Absentees:** Deploy double-touch reminders and direct call protocols for patients with >=2 prior missed visits.
- **4. Telehealth Redirection: Route long-distance patients (>30km) directly to virtual consultations at booking.**

Health Clinic - Executive Attendee Dashboard

51.15%	63.48%	47.54%	61.76%
Overall no-show rate	Lead time > 30 rate	Repeat Absentee Share	Critical Risk Cohort



SLIDE 10 | End-to-End Readiness, Limitations & Closing Remarks

Final Solution Impact, Analytical Limitations, and Key Portfolio Learnings

- **Solution Value:** Establishes a unified data-driven pipeline that recovers lost capacity, lowers costs, and improves patient care access.
- **Analytical Limitations:** Dataset lacks real-time traffic/weather telemetry and socioeconomic income indicators.
- **Key Learnings:** Practical experience in rigorous DAX denominator filtering, cross-track feature engineering, and executive presentation design.
- **Final Status: 100% Tested, Audited, and Finalized for Submission.**

51.15%	63.48%	47.54%	61.76%
Overall No-Show Rate	Lead Time > 30 Rate	Repeat Absentee Share	Critical Risk Cohort